

A Belief-Based Triage Agent for Wearable ECG Alerts: Cost-Asymmetric Decision Making Under Hidden State Uncertainty

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AI Native Shipping Sprint — Week 1
Belief Engine Course

<https://github.com/Julie-max/ai-native-sprint>

Abstract

I built an agent that checks a person's ECG from a wearable device and decides if they need to visit a doctor by reading the ECG ST segment along with prior history and symptoms of the person wearing it. The existing approaches are simple static boundaries for alerts on ECG abnormalities which have high accuracy but miss critical cases. My agent has a cost-based belief system which is updated constantly upon new evidence and discovery; it also holds its decision if there is ambiguity. The thresholds set are: escalate above 35% STEMI probability and dismiss below 0.02% STEMI probability, derived from a cost asymmetry of \$500,000 for a missed critical alert versus \$200 for an unnecessary escalation — a ratio of 2,500:1.

1 Introduction

The problem that exists is that every wearable ECG device out there, and there are many, has simple threshold alarms that treat false electrode artifacts and critical STEMI alerts as the same case. Accuracy as a metric hides this failure: a classifier that is 95% accurate but always wrong on the critical 5% is not 95% useful — it is actively dangerous. My agent works with a cost-based belief system that reasons with given evidence before making a decision, replacing the dumb threshold alarm with a three-action belief engine that distinguishes between hidden states.

2 Agent Design

My four inputs for the agent are ST segment ECG morphology, age, prior medical history, and symptoms experienced. The three hidden states are True STEMI, Hyperventilation, and Electrode Artifact. The three actions are: Escalation with an alert to consult a doctor, Dismiss the alert, or HOLD for further evidence.

The policy is that escalation triggers above 35% STEMI probability, dismissal triggers below 0.02%, and everything in between causes the agent to HOLD for further evidence. Upon new evidence, the belief is updated such that the probabilities change and are re-evaluated against the thresholds.

Threshold Derivation

Thresholds are derived from expected cost calculations:

Threshold	Calculation	Result
Dismiss (0.02%)	$0.0002 \times \$500,000$ vs $0.9998 \times \$200$	$\$100 < \199.96
Escalate (35%)	$0.35 \times \$500,000$ vs $0.65 \times \$200$	$\$175,000 \gg \130

Table 1: Expected cost derivation for action thresholds

Cost Model

Outcome	Cost
Missed true STEMI	\$500,000
Unnecessary escalation	\$200
Cost asymmetry ratio	2,500:1

Table 2: Cost model for the triage agent

3 Experiments

For the experiments, 20 handcrafted test cases have been created, each with a combination for each hidden state. For example, a case with 7% STEMI probability has its belief updated upon feature change until crossing a threshold for a definite action to be taken — starting at HOLD, updating through family history confirmation (12%), symptom onset (28%), and diaphoresis (42%), at which point the agent escalates. Most cases land in HOLD as the agent is conservative by design and appropriately so for this problem statement.

4 Discussion

The discussion on Reddit taught me how the agent I am building is different from a simple wearable device having a dumb threshold — rather, what we have is an agent that updates its belief on new evidence before acting. One limitation is that the test cases are simulated and not drawn from actual clinical recordings. Future directions include using real MIT-BIH Arrhythmia Database data and defining a maximum HOLD time, since HOLD cannot persist indefinitely — if no new evidence arrives within a defined window, the agent escalates by default, because in cardiac events, indefinite waiting is itself a dangerous action.

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References

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